

Gopalakrishnan Thirunellai Venkitachalam

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EDUCATION

Carnegie Mellon University (CMU) (GPA 4.0/4.0)

MS in Mechanical Engineering | **Specialization: Robotics, AI & Motion Planning**

Selected Courses: Planning & Decision-Making for Robotics, Generative AI, Advanced NLP, Intro to Deep Learning

Pittsburgh, PA

May 2026

Indian Institute of Technology Madras (IITM) (GPA 8.71/10.0)

BTech.(Honors) in Mechanical Engineering | Minor in **Artificial Intelligence and Machine Learning**

Selected Courses: Machine Learning, Reinforcement Learning, Robotics, Multi-Armed Bandits, Stochastic Processes

Chennai, India

June 2024

RESEARCH EXPERIENCE

Search Based Planning Lab, Robotics Institute, CMU

Graduate Research Assistant | Advisor: Prof. [Maxim Likhachev](#)

Pittsburgh, PA

Sept. 2024 – Present

- Researching **constant-time motion planning (CTMP)** for constrained robotic manipulation, targeting guaranteed hook–latch disengagement under tight timing budgets.
- Developing a **constraint-aware experience-transfer framework** that reuses a single demonstrated removal motion to generalize across new hook poses, start states, and latch geometries — avoiding full re-planning.
- Prototyping task-space and joint-subspace planning approaches to enable **reliable disengagement under contact constraints**, before hand-off to a downstream CTMP goal-planner.
- Implementing an evaluation stack in **ROS, MoveIt, RViz, and SAPIEN** on a UR10e-class manipulator to benchmark latency, feasibility, and robustness at scale.

PROJECTS

Teaching Assistant — 16-832 Special Topics: Integrated Planning & Learning | CMU

Dec. 2025 – Present

- **Built a reusable planning-and-learning wrapper** around ManiSkill Push-T (Python, SAPIEN), adding deterministic replay, structured planning observations, and controller/state-access utilities to support reproducible constrained-manipulation research and student projects. Github: [PegasusGTV/Planning_wrapper](#)
- **Shipped a from-scratch Multi-Agent Path-Finding (MAPF) environment and benchmark toolkit**, including agent dynamics, collision validation, dataset ingestion, and trajectory visualization — enabling planning-and-learning experimentation without external simulators. Github: [PegasusGTV/mapf](#)

Iterative 2D–3D Inpainting for Single-Image 3D Reconstruction | CMU

Sept. 2025 – Dec. 2025

- **Built an iterative render–inpaint–lift pipeline** combining Stable-Diffusion-based 2D inpainting with Depth-Anything-V2 depth estimation and point-cloud fusion (Poisson surface reconstruction) to recover missing geometry from a single RGB image.
- **Benchmarked against a TripoSR baseline**, improving Multi-View Consistency Score from **0.938** → **0.971** and reducing Chamfer Distance from **0.066** → **0.030** on synthetic multi-view datasets, demonstrating cleaner and more complete reconstructions.

Formation Control and Multi-Agent Pathfinding (MAPF) | CMU

Sept. 2024 – Dec. 2024

- **Developed GIF-PIBT, a formation-aware extension of PIBT** that adds a globally informed formation-cost heuristic (via SVD-based rigid-body alignment) so coordinated swarms can maintain shape while navigating obstacle-dense maps, enabling collision-free planning for multi-agent fleets.
- **Designed a multi-resolution grid and cost-based routing framework** to balance formation preservation vs goal progress, improving routing stability and reducing formation break-up compared to plain PIBT across multiple map environments.

Qualitative Analysis of RL Algorithms in Various Gym Environments | IIT Madras

Jan. 2024 – May 2024

- Built and compared RL agents (SARSA, Q-Learning, Dueling DQN, REINFORCE, Dyna-Q) across Acrobot, CartPole, and Taxi-v3 to study convergence stability, sample-efficiency trade-offs, and failure modes.

SKILLS

Programming: C++, Python, MATLAB

Robotics: ROS1, MoveIt, URDF/Xacro, RViz, Gazebo, ManiSkill, SAPIEN

Planning & Control: Sampling-based Motion Planning (RRT/PRM), MAPF (PIBT / formation control)

Robot Learning / ML: PyTorch, reinforcement learning (value-based & policy-gradient), diffusion-based vision/inpainting

Vision & 3D Perception: OpenCV, depth estimation, point-cloud processing

Tools: Linux, Git, Docker